

Meeting Summary

PA Connect Architecture Recap & Michigan SIM Warehouse Inventory Demo

Attendees

Shaikh Vakeel Shakheer, Periasamy Leena (presenter for warehouse demo), Sarita Tadikamalla, Chirashree Ghosh, Raghavendra Aerpula, Alok Balda (recording)

1. PA Connect Architecture Recap

Shakheer opened the session with a brief recap of the previous day's discussion on PA Connect's project architecture, picking up specifically on upstream and downstream service dependencies.

Background (from prior session)

PA Connect provides secure electronic titling and registration solutions for vehicle dealers. It is a web-based application built on a client-server architecture, handling transactions, inquiries, dealer/DMV reports, report generation, and imports from the DMS. It is the modernized replacement for the earlier Delphi-based product. The tech stack consists of Angular 16 for the front-end UI and .NET 6 APIs for the back end, with Hangfire (within .NET) used for scheduled jobs. Every transaction passes through two back-end systems: after the current host processes it, it is routed to a separate .NET service called Gatekeeper, which handles communication with the DMV; a dedicated .NET team maintains Gatekeeper, so any changes must work across both systems.

Upstream / Downstream Services (today's recap)

- Upstream services: previously discussed on the prior day (the services PA Connect itself depends on / consumes).
- Downstream services: PA Connect has none — no other services depend on or consume PA Connect — so there was nothing further to cover on that point.

No questions were raised before the discussion moved on to the Michigan warehouse/inventory demo.

2. Michigan SIM Warehouse & Inventory Management (Leena Periasamy)

Who Uses It

MADA (Michigan Dealer Association) uses this application to act as the inventory/warehouse management arm on behalf of the Michigan DMV. The DMV supplies this inventory system to them.

Warehouse & Branch Structure

- MADA receives bulk inventory (in the thousands) from the DMV and receives it into the SIM warehouse system.
- Each Michigan DMV branch (e.g., branch 604) corresponds to a different dealer/profile type — dealers, tag agents, financial institutions, etc.
- There are multiple warehouse branches (roughly seven), each tied to a specific profile type; most are used for testing, with limited live inventory in salvage/title service profiles.

- A dedicated CMO account is used for warehouse inventory (e.g., MIWHSE01 in test; a separate one in production), assigned a CVR number, branch, and license number by the DMV.

Order & Fulfillment Workflow

- Customer purchase orders flow into the system daily via an overnight host run; warehouse staff view and fulfill these orders, then ship the inventory to customers.
- Warehouse representatives can also place orders on behalf of new/not-yet-onboarded customer accounts (outside the normal CVR/Michigan Connect/PA Connect workflow), so inventory is ready when onboarding completes.
- Warehouse staff have access to a limited set of menus: view orders, place their own inventory orders, fulfill upcoming orders, and receive purchase orders.
- Additional menus cover on-hand inventory status (available, damaged, missing) and inventory reports (per customer or aggregated for the year).

Placing & Receiving Orders (Live Demo)

- Leena demonstrated creating a purchase order for inventory (example: 'Winter Water Wonderland' plate, quantity 100, PO number MI 89843).
- Orders can be added, edited, deleted, or saved as pending/templates; templates are useful for recurring monthly orders (e.g., decals/stickers).
- When receiving inventory, the correct serial number range must be entered, following the state-specific plate/serial pattern (example range used: 00 to 99). The system flags an error if a range has already been used.
- Received quantity does not need to match the ordered quantity — the system allows partial receipt of on-hand inventory without blocking the user.

Q&A

- Chirashree Ghosh asked whether receiving a partial quantity (e.g., 50 of 100 ordered) creates a separate line per serial number. Leena clarified that a whole range is received as a single line entry, not itemized per serial number. In practice, warehouses often receive very large lots (1,000–4,000+ units) across multiple ranges at once.
- Raghavendra Aerpula asked whether purchase orders originate from the DMV/host directly. Leena clarified that warehouse representatives create purchase orders themselves in the system; they physically receive the inventory from the DMV separately and then log it into the system for tracking and shipping purposes.

Next Steps

The walkthrough was set to continue into Michigan Connect (the customer-facing side) after covering the warehouse representative flow.